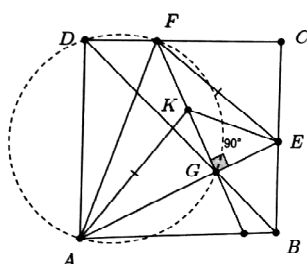


AMTI (NMTC) - 2015
BHASKARA CONTEST - FINAL - JUNIOR LEVEL

1. a) If $p_1, p_2, \dots, p_{2014}$ is an arbitrary rearrangement of $1, 2, 3, \dots, 2014$. Show that

$$\frac{1}{p_1 + p_2} + \frac{1}{p_2 + p_3} + \dots + \frac{1}{p_{2013} + p_{2014}} > \frac{2013}{2016}.$$
 b) Find positive integers n such that $\sqrt{n-1} + \sqrt{n+1}$ is rational.
2. ABCD is a quadrilateral inscribed in a circle of center O. Let BD bisect OC perpendicularly. P is a point on the diagonal AC such that PC = OC. BP cuts AD at E and the circle ABCD at F. Prove that PF is the geometric mean of EF and BF.
3. a) The Fibonacci sequence is defined by $F_0 = 1, F_1 = 1, F_n = F_{n-1} + F_{n-2}, n \geq 2$. Show that $7F_{n+2}^3 - F_n^3 - F_{n+1}^3$ is divisible by F_{n+3} .
 b) If x, y, z are each greater than 1, show that $\frac{x^4}{(y-1)^2} + \frac{y^4}{(z-1)^2} + \frac{z^4}{(x-1)^2} \geq 48$.
4. ABCD is square E and F are points on BC and CD respectively such that AE cuts the diagonal BD at G and FG is perpendicular to AE. K is a point on FG such that AK = EF. Find the measure of the angle EKF.



5. a) If none of a, b, c, x, y, z is zero, and $\frac{x^2(y+z)}{a^3} = \frac{y^2(z+x)}{b^3} = \frac{z^2(x+y)}{c^3} = \frac{xyz}{abc} = 1$ prove that $a^3 + b^3 + c^3 + abc = 0$.
 b) Solve for x, y, z : $\frac{x}{y} + \frac{y}{z} + \frac{z}{x} = \frac{y}{x} + \frac{z}{y} + \frac{x}{z} = x + y + z = 3$.
6. In the dog language BOW, the alphabet consists of the letters B, O, W only. Independently of the choice of the BOW and of length n (i.e.) number of alphabets in the word is n) from which to start, one can construct all the BOW words with length n using iteratively the following rules.
 - i) reverse the order of the letters of the word (if BOWW is a word then if we reverse the order of letters we get WWOBB)
 - ii) replace two consecutive letters as follows :

$$\begin{aligned} BO &\rightarrow WW, & WW &\rightarrow BO, \\ WB &\rightarrow OO, & OO &\rightarrow WB, \\ OW &\rightarrow BB, & BB &\rightarrow OW \end{aligned}$$

Given that BBOWOBOWWOBOWWWOBB is a BOW word, does the BOW language have the following words ?

- a) BWOBWOBWOBWOBOWBOWBOWB
 - b) OBWOBWOBWOBWOBWOBWOBOWBOWBO
7. A merchant bought a quantity of cotton; he exchanged this for oil and he sold the oil. He observed that the number of kg of cotton, the number of liters of oil obtained for each kg and the number of rupees for which he sold formed a decreasing geometric progression. He calculate that if he had obtained 1 kg more of cotton, one liter more of oil for each kg and Rs.1 more for each liter, he would have obtained Rs. 10169 more, whereas if he had obtained one kg less of cotton and one liter less of oil for each kg and Rs. 1 less for each liter, he would have obtained Rs. 9673 less. How much did he actually receive ?

8. There are three towns A, B and C. A person walking from A to B, driving from B to C and riding a horse from C to A completes the journey in $15\frac{1}{2}$ hours. By driving from A to B, riding a horse from B to C and walking from C to A, he could make the journey in 12 hours. On foot he could make the journey in 22 hours, on horseback in $8\frac{1}{4}$ hours and driving in 11 hours. To walk 1 KM, ride 1 KM and drive 1 KM, he takes altogether half an hour. Find the rates at which he travels and the distance between the towns.
